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All points of S have the form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$c_1 = c_2 = 0$
 $\Rightarrow P_0$ is in S !

$\begin{cases} x - 5y + 2z = 9 \\ y = 0 \\ z = 0 \end{cases} \Rightarrow x = 9$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \\ 0 \end{bmatrix} = P_0 + \begin{pmatrix} 9 \\ 0 \\ 0 \end{pmatrix}$

$\Rightarrow x = 9$

Thank you.



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Note on video: In the video above, it looks like the vectors pointing from the space of homogeneous solutions to the parallel plane of inhomogeneous solutions are perpendicular to both planes. This is not in fact the case, and was simply done as a way to make it easier to draw the diagram. (The vectors do point straight up in the standard coordinate system, but the planes should not be drawn as being parallel to the xy -plane.)

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